LASFNet: A Lightweight Attention-Guided Self-Modulation Feature Fusion Network for Multimodal Object Detection

Lei Hao, Lina Xu, *Member, IEEE*, Chang Liu, and Yanni Dong, *Senior Member, IEEE*

***Abstract*—Effective deep feature extraction via feature-level fusion is crucial for multimodal object detection. However, previous studies often involve complex training processes that integrate modality-specific features by stacking multiple feature-level fusion units, leading to significant computational overhead. To address this issue, we propose a lightweight attention-guided self-modulation feature fusion network (LASFNet). The LASFNet adopts a single feature-level fusion unit to enable high-performance detection, thereby simplifying the training process. The attention-guided self-modulation feature fusion (ASFF) module in the model adaptively adjusts the responses of fused features at both global and local levels, promoting comprehensive and enriched feature generation. Additionally, a lightweight feature attention transformation module (FATM) is designed at the neck of LASFNet to enhance the focus on fused features and minimize information loss. Extensive experiments on three representative datasets demonstrate that our approach achieves a favorable efficiency–accuracy tradeoff. Compared to state-of-the-art methods, LASFNet reduced the number of parameters and computational cost by as much as 90% and 85%, respectively, while improving detection accuracy mean average precision (mAP) by 1%–3%. The code will be open-sourced at https://github.com/leileilei2000/LASFNet**

***Index Terms*—Feature fusion, multimodal images, object detection, self-modulation.**

I. Introduction

RECENTLY, significant advancements have been made in object detection technologies based on visible light or infrared imaging [1], [2], [3], [4]. This progress has enabled substantial applications in fields such as remote sensing [5], [6], autonomous driving [7], [8], medical diagnosis [9], [10], traffic management [11], and industrial production [12], [13]. However, the performance of visible light object detection is adversely affected by inadequate lighting conditions and object occlusion. Meanwhile, infrared object detection suffers from issues about thermal source interference and a lack of color information. To address these challenges, researchers have integrated the complementary information of infrared and visible light data to implement multimodal object detection, and this strategy has attracted a positive response and rapid development [14].

In multimodal object detection tasks, features extracted from multiple sensors (e.g., visible light and infrared, visible, and LiDAR) are combined to mitigate the incomplete information of unimodal data. Based on the fusion stage, current multimodal object detection methods can be generally categorized into three types: pixel-level fusion [15], feature-level fusion [16], and decision-level fusion [17]. Pixel-level fusion directly combines multimodal images at the pixel level, generating richer and higher quality fused information to enhance detection accuracy. Decision-level fusion, in contrast, operates after unimodal detectors, aggregating the outputs of multiple sensors to obtain more robust detection results. By contrast, feature-level fusion integrates multimodal features during the feature extraction stage to form a more comprehensive representation, enabling deeper interaction and improved detection performance. Consequently, feature-level fusion has become the most widely adopted fusion strategy in multimodal object detection [18]. However, previous studies often overlook the significant computational overhead associated with feature-level fusion [19], leading to issues such as high power consumption and inference latency. These challenges may hinder the widespread adoption of multimodal detection in embedded systems (e.g., autonomous driving controllers and industrial robots) and resource-constrained edge devices (e.g., smart cameras and drones) [20], [21].

Recently, self-modulation mechanisms have demonstrated remarkable performance in multimodal object detection, as they can adaptively regulate the contribution of each modality's information. Eventually, the most beneficial information for detection is obtained. For example, Yuan and Wei [22] proposed a calibration-complementary mechanism that adaptively adjusts the complementary features between different modalities, improving detection accuracy. Dong et al. [23] and Wang et al. [24] achieved deep fusion by adaptively adjusting modality differences in the hidden state space,

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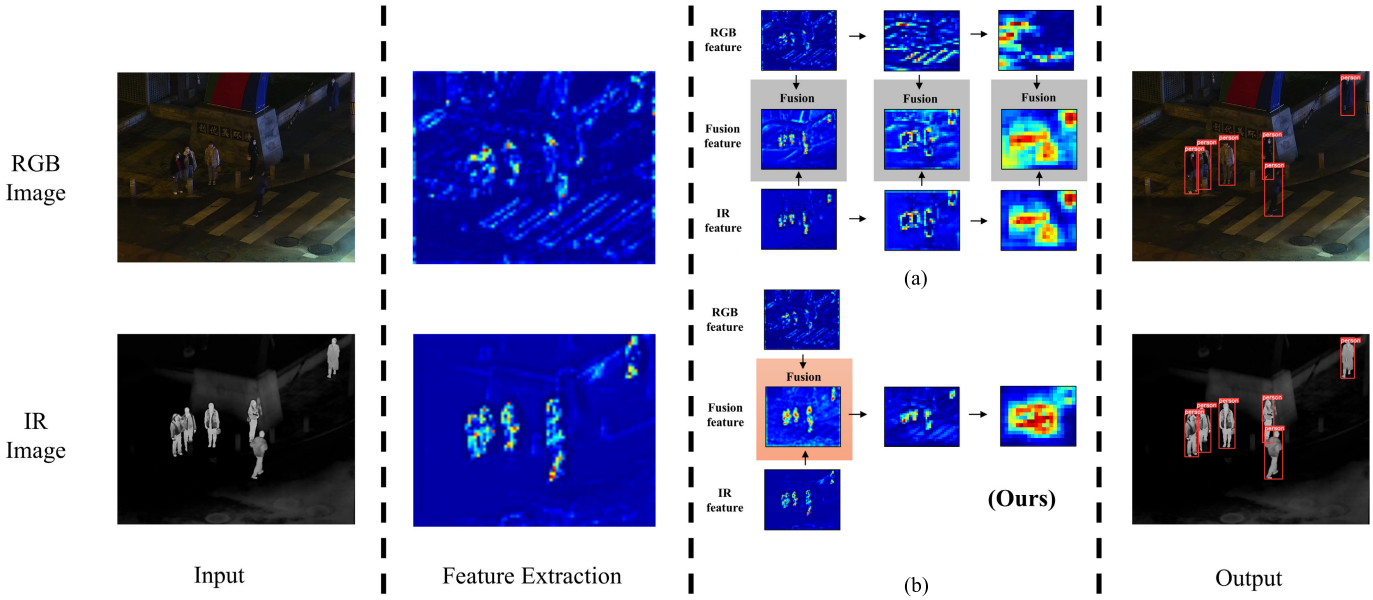


Fig. 1. Popular multimodal fusion baseline and the proposed new baseline. (a) Popular baseline with multifeature fusion units: this approach requires a separate fusion of texture and semantic features from different modalities, which incurs substantial computational overhead to improve detection performance. (b) Proposed new baseline with single-feature fusion units: This method prioritizes the fusion of texture features, followed by the exploration of deeper semantic information within the fused features. Compared to the popular baseline, the proposed baseline achieves equally precise detection performance with a more lightweight and efficient design.

thereby enhancing detection performance. Zhang et al. [25] introduced a multimodal fusion collaborative perception algorithm to enhance the perception capabilities of autonomous vehicles. Jiang et al. [26] achieved adaptive modeling of diverse pedestrian trajectories through a self-attention mechanism learning. Although these methods have shown promising detection performance through self-modulation mechanisms, the substantial computational cost of these models remains a significant concern. Therefore, a key challenge is to develop a low-power feature self-modulation technique that can drive the lightweight evolution of multimodal object detection for real-world applications.

To this end, we propose a lightweight attention-guided self-modulation feature fusion network (LASFNet) for multimodal object detection, which delivers high-performance inference with low computational cost. Unlike previous leveraged multimodal detection baselines [in Fig. 1(a)], the presented new baseline obtains lightweight and efficient detection through only a single feature-level fusion unit [in Fig. 1(b)]. We introduce a novel ASFF module to get rich and robust fused features. The ASFF module performs self-modulation of global and local information guided by attention to facilitate deep fusion. Furthermore, we design a lightweight and flexible feature attention transformation module (FATM) that refines fused features in the deeper layers of the network, effectively mitigating information loss. LASFNet was evaluated on three representative datasets, that is, DroneVehicle [27], LLVIP [28], and VTUAV_det [29]. Experimental results demonstrate that our method achieves a favorable tradeoff between accuracy and efficiency. The main contributions of this article can be summarized as follows.

1) LASFNet is developed for multimodal object detection. By only adopting a single-feature fusion unit, LASFNet maintains accurate detection performance while remaining lightweight.
2) The ASFF module is designed to adaptively fuse global and local features based on modalities' response levels. This module generates rich and robust representations by gradually exploring the information in the fused features from coarse to fine.
3) The FATM is placed in the neck of the network, where it emphasizes key representations in the multimodal fused features, suppresses redundant information, and improves multiscale feature fusion, thereby enhancing detection performance for objects at different scales.

The structure of this article is as follows. Section II briefly reviews related work. Section III details the LASFNet architecture. We present comprehensive experimental results in Section IV. Finally, Section V provides a brief conclusion.

II. Related Work

A. Multimodal Object Detection

Visible light-based object detection serves as a key task in computer vision, and has made substantial progress [32], [33], [34]. However, adverse environmental factors such as cloud cover, fog, and low-light conditions will significantly degrade the quality of visible light images, resulting in suboptimal visual performance of single-modality detection [35]. To address this limitation, infrared data, which complements visible-light images, has been incorporated into object detection. The resulting multimodal object detection framework substantially improves overall detection performance. Zhang et al. [36] employed super-resolution-assisted pixel-level fusion to achieve higher spatial resolution for remote sensing small object detection. However, this approach

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requires significant computational resources, leading to longer processing times. Wang et al. [17] utilized decision-level fusion through multiple branches to mitigate the influence of individual modalities. However, the extensive use of branch structures not only increases computational overhead but may also introduce an imbalance in decision quality. In contrast, feature-level fusion enables flexible and in-depth extraction of rich feature representations, improving overall model performance. Nowadays, the feature-level fusion has been widely adopted in remote sensing object detection. Wang et al. [37] designed a cross-modal enhancement strategy to capture modal correlations and adaptively focus on bimodal features, achieving comprehensive feature generation. Hu et al. [38] proposed a joint infrared-fusion adapter designed to adaptively mitigate feature-level bias for robust tracking performance. Additionally, Liu et al. [39] designed a multimodal complementary information filter to mitigate the impact of feature domain differences between different modalities. Nevertheless, the above methods prioritize high detection accuracy at the expense of increasingly sophisticated fusion modules. The resulting substantial computational cost, which is often overlooked, hinders the broader deployment of multimodal object detection in practice. To address these issues, this article presents the lightweight network LASFNet by introducing an innovative single feature-level fusion strategy that preserves spatial detail perception while achieving simultaneous improvements in accuracy and speed at minimal computational cost, thereby optimizing the balance between efficiency and precision.

B. Self-Modulation in Multimodal Detection

The core idea of self-modulation mechanisms [40] is to enable networks to automatically adjust their internal activations or weights based on input features, thereby better capturing the structure and characteristics of the input data. The transformer [41] captures dependencies between different positions in the input sequence through its self-attention mechanism and dynamically associates information from different modalities, making it particularly well-suited for multimodal information fusion to assist in multimodal object detection. Yuan et al. [42] proposed a complementary fusion transformer that utilizes self-attention to address issues related to misalignment and inaccurate fusion. Helvig et al. [43] combined cross-attention with the transformer-based DINO detector to extract meaningful correlations between modalities. Furthermore, vision Mamba (Vim) achieves the same global visual context modeling capability as the transformer with lower computational complexity, showing significant potential for multimodal object detection [44], [45], [46]. Li et al. [47] introduced a cross-modal fusion Mamba for multispectral object detection under adverse weather conditions. Ren et al. [48] utilized the global perception advantage of Mamba in fusion tasks, combining it with local information to improve the separability of small remote-sensing objects. Given the significant computational overhead introduced by both transformer and Mamba in enhancing multimodal object detection, some researchers have cleverly combined attention information for self-modulation. Chen et al. [49] employed attention and deformable convolutions to adaptively adjust multimodal feature discrepancies, addressing weak misalignment issues in multimodal detection. Berjawi et al. [50] jointly employed frequency-domain filtering and cross-attention mechanisms to enhance multimodal object detection. Although the methods above have proven effective in multimodal object detection tasks, their complexity remains substantial, hindering lightweight requirements. To address this, we propose a lightweight, attention-guided self-modulation feature fusion (ASFF) mechanism that efficiently integrates global and local multimodal information to enhance multimodal detection performance, with a lightweight design.

III. Methodology

A. Architecture Overview

The illustration of LASFNet is depicted in Fig. 2, which comprises three main components: the backbone, the neck, and the detection head. Notably, we introduce two lightweight modules to enhance detection performance, enabling the model to obtain efficient, high-performance detection. First, instead of leveraging multiple parallel feature-level fusion modules in the backbone [51], [52], [53], we design a single feature-level fusion model, ASFF, to progressively and adaptively fuse paired RGB-IR features, yielding a comprehensive and rich fused feature. Second, while maintaining robust multiscale feature extraction through a top–down feature pyramid network (FPN) [30] and bottom–up path aggregation network (PANet) [31] in the neck, we introduce a simple yet flexible FATM to focus on discriminative features and reduce information loss. Finally, the multiscale features from the neck are passed into the detection head to achieve the final detection results.

B. Attention-Guided Self-Modulation Feature Fusion

The proposed ASFF module for comprehensive multimodal fusion through adaptive progressive feature integration. Fig. 3 illustrates the detailed structure of the ASFF module, which comprises three fusion stages: attention fusion, feature modulation fusion, and channel shuffle (CS) fusion.

1) Attention Fusion: In multimodal fusion, attention mechanisms dynamically weight modality regions to focus on key information [54]. In the proposed ASFF module, we first apply a dual-branch structure to the incoming RGB and IR feature maps $\mathbf{P}_{RGB}, \mathbf{P}_{IR} \in \mathbb{R}^{H\times W\times C}$ using a channel attention mechanism (CAM). This mechanism adaptively weights feature channels by importance, generating enhanced multimodal representations $\mathbf{M}_{RGB}, \mathbf{M}_{IR} \in \mathbb{R}^{H\times W\times C}$. The process is defined by

$$\begin{aligned} \mathbf{M}_{IR} &= \mathrm{CAM}\left(\mathbf{P}_{IR}\right) \\ \mathbf{M}_{RGB} &= \mathrm{CAM}\left(\mathbf{P}_{RGB}\right) \\ \mathrm{CAM}\left(x\right) &= x \odot \sigma\left(\mathrm{Conv1d}\left(\mathrm{AP}\left(x\right)\right)\right) \end{aligned} \tag{1}$$

where $\odot$ denotes element-wise multiplication, $\sigma(\cdot)$ represents the sigmoid function, AP$(\cdot)$ and Conv1d$(\cdot)$ correspond to adaptive average pooling and 1-D convolution operations, respectively.

Subsequently, the channel-attended feature maps $\mathbf{M}_{RGB}$ and $\mathbf{M}_{IR}$ are element-wise added to their corresponding original

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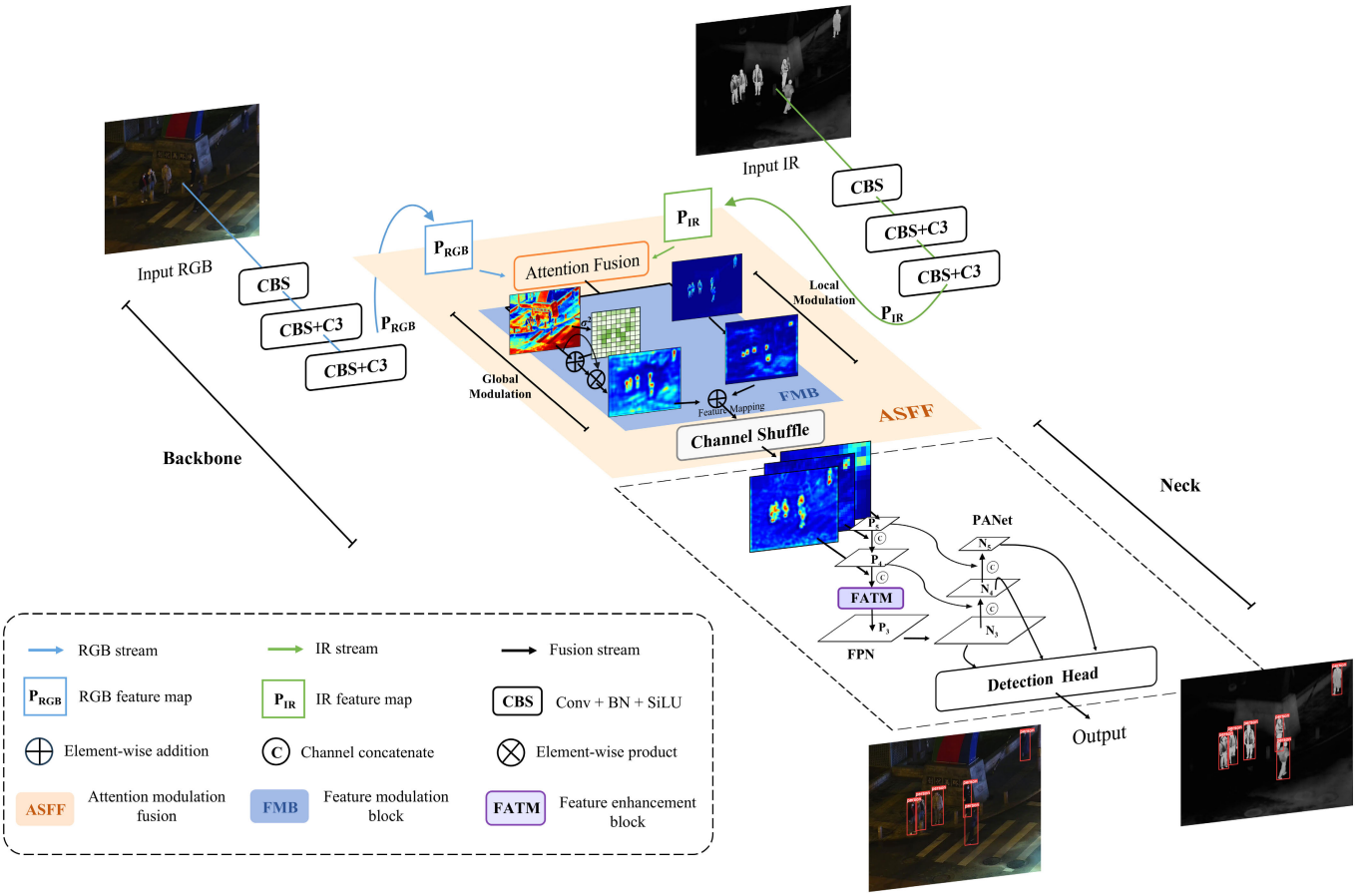


Fig. 2. Overall architecture of LASFNet. The detection network consists of three main components: the feature extraction backbone, the neck, and the detection head. In the feature extraction backbone, the single-feature-level fusion unit, ASFF, is responsible for fusing the P_{RGB} and P_{IR} features extracted by the dual-stream structure. In the neck, we retain the FPN [30] with a top–down structure and the PANet [31] with a bottom–up structure, while introducing the FATM to enhance the network's ability to focus on critical features, thereby reducing feature loss.

inputs $\mathbf{P}_{RGB}$ and $\mathbf{P}_{IR}$, forming residual connections. This operation retains the original spatial and channel information while incorporating the attention-adjusted weights, enhancing the model's focus on important features. The resulting sums are then passed through depthwise convolutions to preserve the independence of the channel features, yielding enhanced modality feature maps $\hat{\mathbf{M}}_{RGB}, \hat{\mathbf{M}}_{IR} \in \mathbb{R}^{H\times W\times C}$, as defined by

$$\begin{aligned}\hat{\mathbf{M}}_{IR} &= \mathrm{DWConv}\left(\mathbf{P}_{IR} + \mathbf{M}_{IR}\right)\\ \hat{\mathbf{M}}_{RGB} &= \mathrm{DWConv}\left(\mathbf{P}_{RGB} + \mathbf{M}_{RGB}\right)\end{aligned} \tag{2}$$

where DWConv denotes depthwise convolutions with a 3×3 kernel. We fuse the enhanced RGB and IR feature maps $(\hat{\mathbf{M}}_{RGB}, \hat{\mathbf{M}}_{IR})$ to improve cross-modal complementarity via element-wise addition. A positional attention mechanism (PAM) then spatially weights features along horizontal/vertical axes, amplifying discriminative local features. The final attention-fused feature map $\mathbf{F}_a$ is computed as

$$\begin{aligned}\mathbf{F}_a &= \mathrm{PAM}\left(\hat{\mathbf{M}}_{RGB} + \hat{\mathbf{M}}_{IR}\right)\\ \mathrm{PAM}\left(x\right) &= x \odot \mathrm{Att}_h\left(x\right) \odot \mathrm{Att}_v\left(x\right)\\ \mathrm{Att}_i\left(x\right) &= \sigma\left(\mathrm{Conv}\left(\mathrm{Avg}_i\left(x\right)\right)\right)\end{aligned} \tag{3}$$

where $\mathrm{Att}_h(\cdot)$ and $\mathrm{Att}_v(\cdot)$ compute the horizontal and vertical position attention weights by applying average pooling $\mathrm{Avg}_h(\cdot)$ along the width and $\mathrm{Avg}_v(\cdot)$ along the height of the input, respectively. $\mathrm{Conv}(\cdot)$ represents a 1×1 2-D convolutional layer.

2) Feature Modulation Fusion: The feature modulation block (FMB) synergizes dynamic feature modulation (DFM) and feature mapping (FM). This architecture strengthens multiscale representations and optimizes feature learning in attention-fused patterns. Specifically, DFM adapts the input features through dynamic modulation to better capture the relationships between global and local features. The process begins by normalizing the attention-fused features $\mathbf{F}_a$, followed by a 1×1 convolution to expand the channels. The channels are then split into two branches for further processing. This operation can be mathematically expressed as

$$\{\mathbf{X}, \mathbf{Y}\} = S\left(\mathrm{Conv}\left(\|\mathbf{F}_a\|_2\right)\right) \tag{4}$$

where $\|\cdot\|_2$ denotes L2 normalization, and $S(\cdot)$ refers to the channel splitting operation. Then, the global and local branches process $\mathbf{X}$ and $\mathbf{Y} \in \mathbb{R}^{H\times W\times C}$ in parallel to generate global features $\mathbf{X}_g$ and local features $\mathbf{Y}_l$.

In the global branch, we first apply adaptive max pooling to downsample the input and obtain salient representations, then feed it into a 3×3 DWConv to capture the global structural

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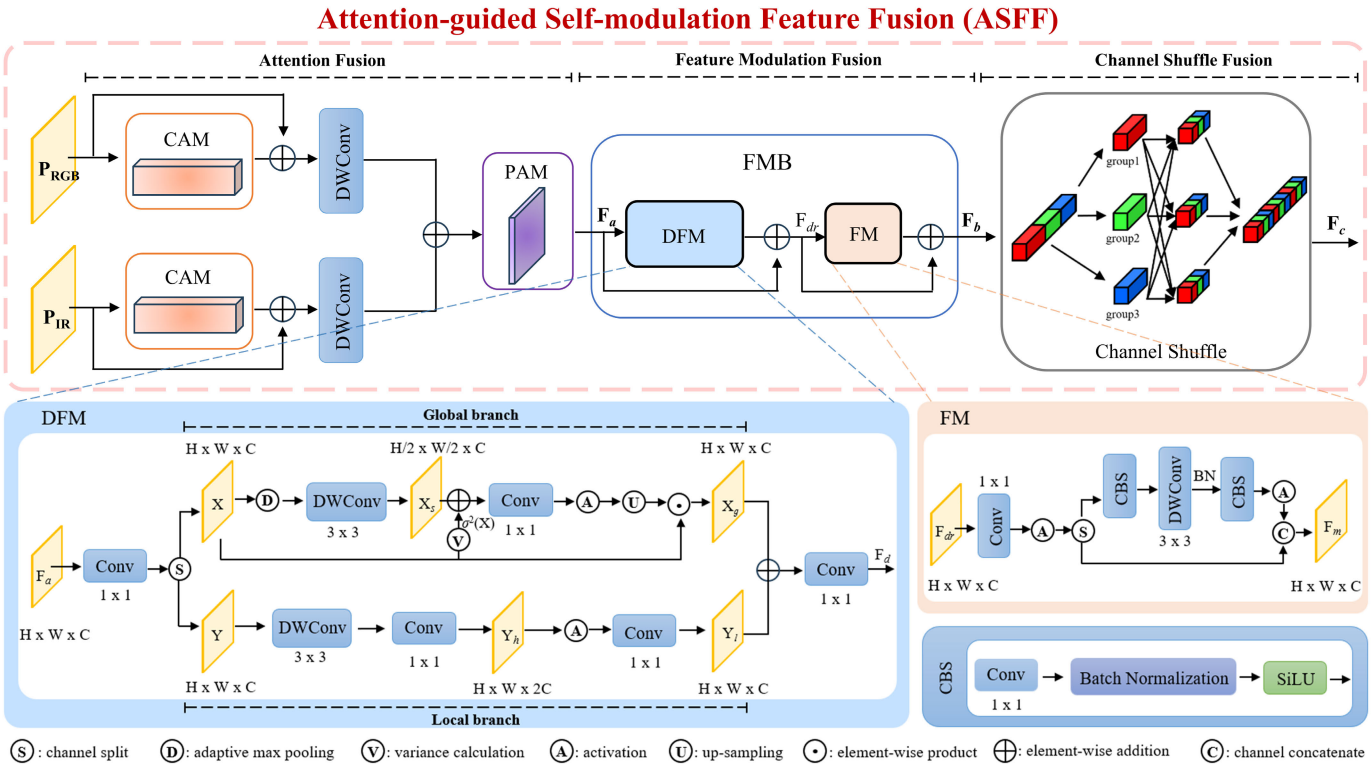


Fig. 3. Detailed architecture of our proposed ASFF module. The fusion process is divided into three stages: attention fusion, feature modulation fusion, and channel shuffling fusion. The RGB and IR feature maps are the inputs of the whole fusion process, and the outputs of the three stages are $\mathbf{F}_a$, $\mathbf{F}_b$, and $\mathbf{F}_c$.

information $\mathbf{X}_s \in \mathbb{R}^{H/2\times W/2\times C}$

$$\mathbf{X}_s = \mathrm{DWConv}\left(D\left(\mathbf{X}\right)\right) \tag{5}$$

where $D(\cdot)$ denotes the adaptive max pooling operation with a scale factor of 2. To embed the global feature information and adaptively adjust the global structural information $\mathbf{X}_s$, we introduce the variance $\sigma^2(\mathbf{X})$ to provide spatial variability for feature fusion, helping the model focus on feature variations in different regions. Two learnable parameters α and β allow the model to dynamically adjust the contributions of the variance and $\mathbf{X}_s$, enabling adaptive fusion of information from different spatial regions. The result is then merged using a 1×1 convolution

$$\begin{aligned}\mathbf{X}_m &= \mathrm{Conv}\left(\mathbf{X}_s \cdot \alpha + \sigma^2\left(\mathbf{X}\right) \cdot \beta\right)\\ \sigma^2\left(\mathbf{X}\right) &= \frac{1}{N}\sum_{i=0}^{N-1}\left(x_i - \mu\right)^2\end{aligned} \tag{6}$$

where $\sigma^2(\mathbf{X}) \in \mathbb{R}^{1\times 1\times C}$ is the variance of $\mathbf{X}$. N is the total number of pixels, x_i represents each pixel value, and μ is the mean of all pixel values. $\mathbf{X}_m \in \mathbb{R}^{H\times W\times C}$ represents the globally modulated feature. Finally, the modulated feature $\mathbf{X}_m$ is used to adjust the original input $\mathbf{X}$ globally, producing the representative global branch feature $\mathbf{X}_g \in \mathbb{R}^{H\times W\times C}$

$$\mathbf{X}_g = \mathbf{X} \odot U\left(\phi\left(\mathbf{X}_m\right)\right) \tag{7}$$

where $\phi(\cdot)$ represents the GELU activation function, $U(\cdot)$ denotes the nearest-neighbor upsampling operation.

Local detail features complement the global features to provide more contextual information and enhance the model's expressiveness. We design a simple bottleneck-inspired local branch to capture local features. Specifically, we first perform a dimensionality increase using a 3×3 depthwise convolution to encode local information in Y, resulting in $\mathbf{Y}_h$. Then, we apply 1×1 convolutions with GELU activations to reduce the dimensionality and enhance the local features, $\mathbf{Y}_l \in \mathbb{R}^{H\times W\times C}$, as follows:

$$\mathbf{Y}_h = \mathrm{Conv}\left(\mathrm{DWConv}\left(\mathbf{Y}\right)\right) \tag{8}$$

$$\mathbf{Y}_l = \mathrm{Conv}\left(\phi\left(\mathbf{Y}_h\right)\right) \tag{9}$$

where $\mathbf{Y}_h \in \mathbb{R}^{H\times W\times 2C}$ represents the locally encoded information after dimensionality expansion. Finally, the global and local features are fused through element-wise addition and passed through a 1×1 convolution to produce the representative output of the DFM module $\mathbf{F}_d \in \mathbb{R}^{H\times W\times C}$, as follows:

$$\mathbf{F}_d = \mathrm{Conv}\left(\mathbf{X}_g + \mathbf{Y}_l\right) \tag{10}$$

where $\mathbf{X}_g$ and $\mathbf{Y}_l$ are the global and local features, respectively.

The FM module further refines the output features of the DFM module through different channel-wise processing, enhancing the ability to capture both global and local features while avoiding excessive computational overhead. As shown in Fig. 3, the FM module first employs a 1×1 convolution with GELU activation on the residual connection feature $\mathbf{F}_{dr}$ to perform dimensionality expansion, thereby increasing the feature channels to facilitate information interaction. The expanded features are then split into two parts, $\{\mathbf{F}_1, \mathbf{F}_2\}$, the features with fewer channels, $\mathbf{F}_1$, undergo local encoding, while $\mathbf{F}_2$ remains unchanged to preserve more global information. The processed

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$\mathbf{F}_1$ and $\mathbf{F}_2$ are concatenated and passed through a final 1×1 convolution to fuse the features and obtain the representative output $\mathbf{F}_m$, restoring the dimensions to the original input state. This process is described by

$$\begin{aligned} \{\mathbf{F}_1, \mathbf{F}_2\} &= S\left(\phi\left(\text{Conv}\left(\|\mathbf{F}_{dr}\|_2\right)\right)\right) \\ \mathbf{F}_m &= \text{Conv}\left(\text{Concat}\left[\phi\left(W\left(\mathbf{F}_1\right)\right), \mathbf{F}_2\right]\right) \\ W(x) &= (\text{CBS}\left(\text{BN}\left(\text{DWConv}\left(\text{CBS}\left(x\right)\right)\right)\right)) \end{aligned} \tag{11}$$

where $\mathbf{F}_1 \in \mathbb{R}^{H\times W\times C/2}$, $\mathbf{F}_2 \in \mathbb{R}^{H\times W\times 3C/2}$, $W(\cdot)$ denotes the local encoding operation applied to $\mathbf{F}_1$, and Concat$(\cdot)$ denotes channel concatenation. CBS$(\cdot)$ refers to a sequence of 1×1 convolution, normalization, and SiLU activation, while BN$(\cdot)$ denotes the batch normalization operation. As described earlier, the DFM and FM are combined into a FMB for a second feature fusion, generating a rich representation that combines spatial, channel, and both global and local multiscale features. To stabilize model training and retain more informative features, residual connections are incorporated in the FMB. The final output $\mathbf{F}_b$ of the FMB module is computed as

$$\begin{aligned} \mathbf{F}_b &= \text{FM}\left(\mathbf{F}_{dr}\right) + \mathbf{F}_{dr} \\ \mathbf{F}_{dr} &= \text{DFM}\left(\mathbf{F}_a\right) + \mathbf{F}_a. \end{aligned} \tag{12}$$

3) CS Fusion: CS [55], [56] can be viewed as a lightweight channel reorganization operation. In traditional convolutional neural networks, convolution operations extract local features through sliding windows in the image space. However, interactions between channels are typically limited. The CS operation breaks the independence between channels by shuffling the order of channels across different groups, enabling the network to exchange information between multiple channel groups. This approach promotes the complementarity between different features at a low computational cost, thus improving the model's representational capacity. The features $\mathbf{F}_c$ obtained after channel shuffling, can be computed as follows:

$$\begin{aligned} \mathbf{x}' &= \text{reshape}\left(\mathbf{x}, B, G, C/G, H, W\right) \\ \mathbf{x}'' &= \text{transpose}\left(\mathbf{x}', 1, 2\right) \\ \mathbf{F}_c &= \text{reshape}\left(\mathbf{x}'', B, C, H, W\right) \end{aligned} \tag{13}$$

where the reshape operation refers to a feature reshaping process, $\mathbf{x} \in \mathbb{R}^{B\times C\times H\times W}$, with B representing the batch size, G being the number of channel groups, and C/G denoting the number of channels per group. The reshaped tensor $\mathbf{x}'$ has the shape $(B, G, C/G, H, W)$ which can be interpreted as dividing C channels into G groups, each containing C/G channels. The transpose operation swaps the group dimension G with the channel dimension C/G, thus enabling the elements of each channel group to be distributed across different groups, increasing the interaction between channels. After reshaping back to the original format, the final output of the CS fusion is obtained as $\mathbf{F}_c$. Algorithm 1 shows the process involved in the ASFF module.

ASFF reduces computational overhead with an efficient feature fusion strategy, using attention weighting and dynamic modulation to retain key features, thereby enhancing multimodal feature learning. This lightweight design offers significant advantages in multimodal object detection, especially in embedded or resource-constrained environments, balancing computational efficiency and performance.

Algorithm 1 ASFF Module

Input: Pairwise visible/infrared features: $\mathbf{P}_{RGB}$, $\mathbf{P}_{IR}$
Output: Fusion feature $\mathbf{F}_c$
Stage 1: Attention Fusion
1. Obtain channel-weighted features via channel attention: $\mathbf{M}_{RGB}, \mathbf{M}_{IR} \leftarrow \text{CAM}(\mathbf{P}_{RGB}), \text{CAM}(\mathbf{P}_{IR})$
2. Enhance features via residual connection: $\hat{\mathbf{M}}_{RGB} \leftarrow \text{DWConv}(\mathbf{M}_{RGB} + \mathbf{P}_{RGB})$ $\hat{\mathbf{M}}_{IR} \leftarrow \text{DWConv}(\mathbf{P}_{IR} + \mathbf{M}_{IR})$
3. Generate attention fusion feature: $\mathbf{F}_a \leftarrow \text{PAM}(\hat{\mathbf{M}}_{RGB} + \hat{\mathbf{M}}_{IR})$
Stage 2: Feature Modulation Fusion
4. Split $\mathbf{F}_a$ into two branches: Global branch: $\mathbf{X}_g \leftarrow \text{GlobalProcess}(\mathbf{X})$ Local branch: $\mathbf{Y}_l \leftarrow \text{LocalProcess}(\mathbf{Y})$
5. Fuse two branches: $\mathbf{F}_d \leftarrow \text{Conv}(\mathbf{X}_g + \mathbf{Y}_l)$
6. Obtain modulated feature via residual: $\mathbf{F}_{dr} \leftarrow \mathbf{F}_d + \mathbf{F}_a$ $\mathbf{F}_b \leftarrow \text{FM}(\mathbf{F}_{dr}) + \mathbf{F}_{dr}$
Stage 3: Channel Shuffle Fusion
7. Reshape feature: $\mathbf{x}' \leftarrow \text{reshape}(\mathbf{F}_b)$
8. Transpose for interaction: $\mathbf{x}'' \leftarrow \text{transpose}(\mathbf{x}')$
9. Recover shape: $\mathbf{F}_c \leftarrow \text{reshape}(\mathbf{x}'')$
return $\mathbf{F}_c$

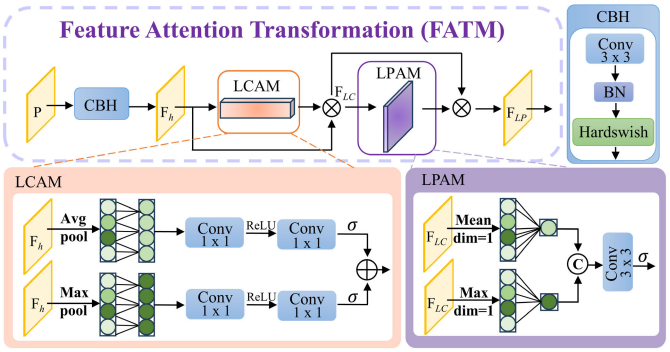


Fig. 4. Detailed architecture of our proposed FATM. The LCAM and the LPAM, synergistically preserve critical feature representations while effectively mitigating information degradation throughout the process.

C. Feature Attention Transformation Module

Multistage sampling and fusion in the neck may blur backbone-extracted features while introducing redundancy and noise, degrading detection quality. To address this, we propose a lightweight FATM at the end of FPN. By applying channel and positional weighting, FATM suppresses irrelevant information while preserving critical features. This enhancement helps subsequent PAN networks better integrate multiscale features, ultimately improving detection performance.

In practice, the FATM first processes the input feature tensor $\mathbf{P}$ through a conv-bn-Hardswish (CBH) block. As shown in Fig. 4, this block employs a standard 3×3 convolution to

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effectively capture spatial features, while the batch normalization and Hardswish activation are employed to enhance the training stability and nonlinear representation capability. The operation is formulated as

$$\mathbf{F}_h = \text{Hardswish}\left(\text{BN}\left(\text{Conv}_{3\times3}\left(\mathbf{P}\right)\right)\right) \tag{14}$$

where Hardswish(·) represents the Hardswish nonlinear activation function, and $\text{Conv}_{3\times3}(\cdot)$ represents a 3×3 2-D convolutional layer. The resulting feature map $\mathbf{F}_h$ is then fed into the lightweight channel attention module (LCAM) to recalibrate channel-wise importance dynamically, which promote the network to focus on more informative features. The channel-attention-enhanced feature $\mathbf{F}_{LC}$ is given by

$$\mathbf{F}_{LC} = \mathbf{F}_h \odot \text{LCAM}\left(\mathbf{F}_h\right) \tag{15}$$

where the LCAM(·) extracts attention information from the global average and local maximum responses of the input features using a dual-branch structure. The final channel attention weights are obtained by fusing this information, and the operation is expressed as

$$\begin{aligned}\text{LCAM}\left(\mathbf{F}_h\right) = {} & \sigma\left(\text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\text{AP}\left(\mathbf{F}_h\right)\right)\right)\right)\right) \\ & + \sigma\left(\text{Conv}\left(\text{ReLU}\left(\text{Conv}\left(\text{MP}\left(\mathbf{F}_h\right)\right)\right)\right)\right)\end{aligned} \tag{16}$$

where $\sigma(\cdot)$ denotes the sigmoid function, and ReLU(·) represents the ReLU activation function. AP(·) and MP(·) refer to adaptive average pooling and adaptive max pooling, respectively. After applying channel attention, the weighted features are passed into the lightweight positional attention module (LPAM), which adaptively focuses on important spatial regions, thus enabling dynamic adjustment of the positional information in the feature map. The positional attention-enhanced features $\mathbf{F}_{LP}$ is represented as the final output as follows:

$$\mathbf{F}_{LP} = \mathbf{F}_{LC} \odot \text{LPAM}\left(\mathbf{F}_{LC}\right) \tag{17}$$

where LPAM(·) performs global average pooling and global max pooling along the channel dimension in a dual-branch manner, followed by fusion to obtain the final position attention weights, which can be expressed as

$$\text{LPAM}\left(\mathbf{F}_{LC}\right) = \sigma\left(\text{Conv}_{3\times3}\left(\text{Concat}\left[\mathbf{F}_{Max}, \mathbf{F}_{Mean}\right]\right)\right) \tag{18}$$

where $\mathbf{F}_{Max}$ denotes the result of applying global max pooling along the channel dimension of $\mathbf{F}_{LC}$, while $\mathbf{F}_{Mean}$ denotes the result of applying global average pooling along the channel dimension of $\mathbf{F}_{LC}$. Algorithm 2 shows the process involved in the FATM module.

The FATM module achieves a dual objective of lightweight design and effective feature enhancement by jointly leveraging channel and positional attention mechanisms. It substantially strengthens the model's ability to selectively focus on key features while maintaining low computational overhead and a reduced number of parameters.

D. Loss Function

The overall loss of the model remains consistent with the total loss used in YOLOv5 [57], consisting of three components: the bounding box regression loss, object classification loss, and object confidence loss. The final loss is expressed as

$$L_{\text{total}} = \lambda_{\text{box}} \cdot L_{\text{box}} + \lambda_{\text{obj}} \cdot L_{\text{obj}} + \lambda_{\text{cls}} \cdot L_{\text{cls}} \tag{19}$$

where the hyperparameters λ_{box}, λ_{obj}, and λ_{cls} are used to adjust the relative importance of each loss component.

Algorithm 2 FATM Module

Input: Feature map $\mathbf{P}$
Output: Enhanced feature $\mathbf{F}_{LP}$
1. Extract features via CBH:
   $\mathbf{F}_h \leftarrow \text{CBH}(\mathbf{P})$
2. Dual-branch channel attention LCAM:
   AAP branch: $\mathbf{W}_g \leftarrow \sigma(\text{CRC}(\text{AP}(\mathbf{F}_h)))$
   AMP branch: $\mathbf{W}_l \leftarrow \sigma(\text{CRC}(\text{MP}(\mathbf{F}_h)))$
3. Fuse channel weights:
   $\mathbf{F}_{LC} \leftarrow \mathbf{F}_h \odot (\mathbf{W}_g + \mathbf{W}_l)$
4. Dual-branch position attention LPAM:
   GAP branch: $\mathbf{S}_g \leftarrow \text{GAP}(\mathbf{F}_{LC})$
   GMP branch: $\mathbf{S}_m \leftarrow \text{GMP}(\mathbf{F}_{LC})$
5. Fuse position weights:
   $\mathbf{F}_{LP} \leftarrow \mathbf{F}_{LC} \odot \sigma(\text{Conv}_{3\times3}(\text{Concat}[\mathbf{S}_g, \mathbf{S}_m]))$
   return $\mathbf{F}_{LP}$

IV. Experiments

A. Datasets

To ensure a comprehensive and fair evaluation, we conduct experiments on three representative and widely used multimodal object detection benchmarks. The benchmarks are DroneVehicle, LLVIP, and VTUAV_det.

1) DroneVehicle [27]: This dataset is primarily designed for visible-infrared (RGB-IR) modality-based aerial vehicle detection, and includes images captured at various times throughout the day. The objects in this dataset are relatively small, and challenges include poor object visibility in the visible modality during late-night hours, as well as thermal artifacts resembling vehicles in the infrared modality. The dataset consists of five object classes, with a total of 19 459 RGB-IR image pairs, of which 17 990 pairs are used for training and the remaining 1469 pairs are used for testing. Due to the substantial differences between the visible and infrared modalities, this dataset is particularly valuable for evaluating the model's performance in complementary data fusion.

2) LLVIP [28]: This dataset is a large-scale pedestrian detection dataset captured under low-light conditions using a binocular camera. It contains 15 488 pairs of strictly aligned RGB-IR images, with 12 025 pairs used for training and 3463 pairs used for testing. The dataset has undergone rigorous alignment, and the object scales in the images are relatively consistent compared to the other two datasets, with a generally larger object scale. This makes it suitable for assessing the model's overall detection performance.

3) VTUAV_Det: This dataset is derived from the VTUAV [58] dataset, which was originally created for large-scale multimodal drone tracking, and has been reannotated by

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Zhang et al. [29] for multimodal object detection. It includes 11 392 image pairs used for training, with 5378 image pairs reserved for testing. The dataset contains objects of three different sizes (small, medium, and large). Moreover, the images are not strictly aligned, and the dataset contains challenges such as significant object occlusion and image blur. Therefore, this dataset is highly suitable for evaluating the model's performance in handling challenges related to multiscale variations, object occlusion, and image alignment.

B. Implementation Details

The proposed method introduces a novel backbone built upon CSPDarknet53 from YOLOv5 in PyTorch. After extracting feature maps from each branch in the dual-branch architecture, a single fusion unit is performed. The model is trained using the SGD optimizer on two NVIDIA GeForce RTX 3090 GPUs. When training on the DroneVehicle dataset, we converted the original oriented bounding boxes to horizontal bounding boxes to ensure a unified standard for experimentation. The network was trained with a batch size of 48, an initial learning rate of 0.01, momentum set to 0.937, and weight decay set to 0.0005, with a total of 300 epochs. When evaluating inference speed, all methods were run on a single GPU with a batch size of 1.

C. Accuracy Metrics

We evaluate the proposed method using commonly adopted detection metrics in the field of object detection, including mean average precision (mAP) and mAP50, along with giga floating-point operations per second (GFLOPs), parameter size (Params), and testing time for comprehensive model assessment. The mAP is a key metric for evaluating the detection performance of a model. It measures the precision of an object detection model across multiple classes and different intersection over union (IoU) thresholds, providing a comprehensive and rigorous evaluation standard. Specifically, mAP50 refers to the mAP computed at an IoU threshold of 0.5, which is typically used as a baseline evaluation metric for detection tasks. GFLOPs is a crucial indicator for assessing the computational demand and efficiency of a model during inference. Params refers to the storage requirement of the model, serving as an essential metric for evaluating the model's feasibility and deployment challenges on storage-constrained devices. The testing time, measured in milliseconds (ms), reflects the actual inference speed by indicating the time required for the model to process a pair of images, which is critical for real-world applications.

D. Loss Component Analysis

A loss component during training fundamentally ensures the generation of corresponding high-quality output information during inference. To validate the gain of each loss component in the YOLOv5 detector and understand its contribution to the final model performance, we conducted a diagnostic analysis by suppressing specific output channels at inference time. We simulate the effect of ablating the corresponding component, thereby inversely verifying the value of each loss function.

TABLE I
PERFORMANCE UNDER DIFFERENT OUTPUT CONFIGURATIONS AT INFERENCE TIME ON THE LLVIP DATASET

Boxes	Objectness	Classification	mAP	Testing time(ms)
✓	×	×	0.0381	139.9
×	✓	×	0.215	73.9
×	×	✓	0.0003	138.7
✓	✓	×	0.652	68.6
✓	×	✓	0.0381	133.5
×	✓	✓	0.228	72.9
✓	✓	✓	**0.676**	**49.8**

Specifically, we utilized a single model weight trained with the complete loss function. During testing, we modified the forward pass output to simulate scenarios lacking outputs associated with different losses. For instance, simulating the absence of the objectness loss L_{obj} by outputting only bounding boxes, without objectness scores, and simulating the absence of the classification loss L_{cls} by outputting bounding boxes and objectness scores, but without classification scores.

The results are presented in Table I. We observe that the completeness of the output information has a decisive impact on both detection accuracy and inference speed.

1) *Impact on Accuracy:* When the output information is incomplete (e.g., lacking classification scores), the mAP has a plunge, as seen when comparing the second row with the sixth row, and the fourth row with the seventh row. This is because it hinders the model's ability to perform effective category discrimination.
2) *Impact on Speed:* More importantly, the output information directly determines the computational load of the postprocessing stage. When the system can only obtain bounding boxes without objectness scores (first row of Table I, the subsequent nonmaximum suppression (NMS) step must process a massive number of candidate boxes due to the inability to perform prefiltering, resulting in the slowest inference speed. Conversely, when objectness scores are available (fourth row of Table I, a large number of low-confidence proposals can be quickly filtered out, significantly reducing the computational burden on NMS and thus achieving the highest inference efficiency.

Analysis confirms that the most efficient model is achieved when the detection head provides complete output information. Therefore, the complete YOLOv5 is adopted in subsequent experiments. This guarantees the robustness of the baseline performance and ensures that any performance gains can be attributed to our proposed components.

E. Ablation Experiments

In this section, we evaluate the effectiveness of the proposed LASFNet and perform an ablation study by configuring the proposed modules in different ways. As the LLVIP has undergone rigorous alignment and provides higher image resolution, it ensures the reliability of model performance evaluation

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TABLE II
COMPARISON OF THE PERFORMANCE OF THE MULTIFEATURE FUSION BASELINE AND THE PROPOSED SINGLE-FEATURE FUSION BASELINE ON THE LLVIP DATASET

Method	mAP50	mAP	Params (M)	GFLOPs
Multi Fusion Baseline	**0.965**	0.643	11.2	26.1
Single Fusion Baseline	0.964	**0.644**	**7.2**	**20.8**

TABLE III
PERFORMANCE COMPARISON OF ASFF MODULE WITH DIFFERENT COMPONENTS ON THE LLVIP DATASET

Att	FMB	CS	mAP50	mAP	Params (M)	GFLOPs
×	×	×	0.964	0.644	**7.2**	**20.8**
✓	×	×	0.965	0.646	**7.2**	**20.8**
×	✓	×	0.965	0.648	7.5	23.7
×	×	✓	0.966	0.646	**7.2**	**20.8**
✓	✓	×	0.965	0.659	7.5	23.7
×	✓	✓	0.967	0.649	7.5	23.7
✓	×	✓	0.967	0.652	**7.2**	**20.8**
✓	✓	✓	**0.969**	**0.667**	7.5	23.7

across different configurations. Therefore, we chose the LLVIP for the ablation study. First, to assess the feasibility of the single-feature fusion approach, we compare the detection performance of the popular multifeature fusion baseline and the proposed single-feature fusion baseline. The two baseline fusion methods are shown in Fig. 1. Additionally, we investigate the performance improvements brought by the proposed ASFF and FATM modules within the baseline network.

1) Effectiveness of the Proposed Baseline: We compare the detection results and model performance of two fusion baselines in Table II, where both baseline methods use element-wise addition for fusion. In terms of detection accuracy, the performance of the single-feature fusion baseline is comparable to that of the multifeature fusion baseline. However, in terms of model efficiency, the single-feature fusion baseline reduces the number of Params by approximately 36% and reduces GFLOPs by 20% compared to the multifeature fusion baseline. Therefore, the proposed single-feature fusion baseline demonstrates equivalent detection performance with a more efficient design, validating its effectiveness as a new baseline for multimodal detection.

2) Effectiveness of the ASFF: We evaluate the performance of the three components of the ASFF module: attention fusion, FMB, and CS fusion. Table III presents a performance comparison of different component combinations, with the first row showing the baseline results without ASFF and the last row displaying the complete ASFF results. In the absence of attention fusion, we replace it with direct element-wise summation. As shown in the first four rows of Table III, the application of each individual fusion component significantly improves detection performance, indicating that all three components contribute to multimodal information fusion.

The experimental results in the last four rows of Table III depict that pairwise combinations of Att, FMB, and CS lead to better performance than using each component individually.

TABLE IV
PERFORMANCE COMPARISON OF LASFNET WITH DIFFERENT COMPONENTS ON THE LLVIP DATASET

ASFF	FATM	mAP50	mAP	Params (M)	GFLOPs
×	×	0.964	0.644	**7.2**	**20.8**
×	✓	0.965	0.657	7.5	23.6
✓	×	0.969	0.667	7.5	23.7
✓	✓	**0.974**	**0.676**	7.7	26.6

When all three components are combined into the complete ASFF module, the mAP improves by 2.3% compared to the baseline. This demonstrates that the three fusion components complement each other, effectively enhancing the extraction of fused information and generating comprehensive and rich representative features. Attention fusion guides the model to focus on the most valuable information from different modalities in terms of both channels and spatial locations, providing strongly correlated input for the FMB. After applying Att, FMB effectively adapts and adjusts the fused information at both global and local feature levels, facilitating multimodal fusion. Meanwhile, CS enhances the interaction of fused information without adding computational burden, allowing the network to extract meaningful features from multimodal data better. These results fully validate the lightweight and effective nature of the proposed ASFF for multimodal feature fusion.

3) Effectiveness of FATM: This section examines the contribution of the FATM to model performance. We compare four experimental configurations: the first two rows of Table IV show the results with and without the FATM module in the proposed baseline, and the last two rows present the results when the FATM module is either removed or retained in the baseline framework with the ASFF module. In the absence of ASFF, element-wise summation was used as an alternative. From the comparison of the first two rows, it is clear that adding the FATM module to the baseline increases the overall mAP by 1.3% with a minimal increase in parameters. Similarly, the analysis of the last two rows demonstrates that incorporating the FATM module into the ASFF-based fusion framework improves the overall mAP by nearly 1.0%, while maintaining a lightweight architecture. These results confirm that the proposed FATM module effectively enhances the model's multiscale detection performance in a lightweight manner.

F. Comparison With State-of-the-Art Methods

In this section, we compare the proposed method with several state-of-the-art approaches on three datasets: DroneVehicle, LLVIP, and VTUAV_det. The comparative results are summarized in Tables V–VII. Our method consistently exhibits a better tradeoff between lightweight design and high detection accuracy across all datasets. Although our model contains slightly more parameters than SuperYOLO and GHOST, it significantly reduces computational complexity while improving the mAP by 2%–5%. Compared to other methods, LASFNet reduces the number of parameters by

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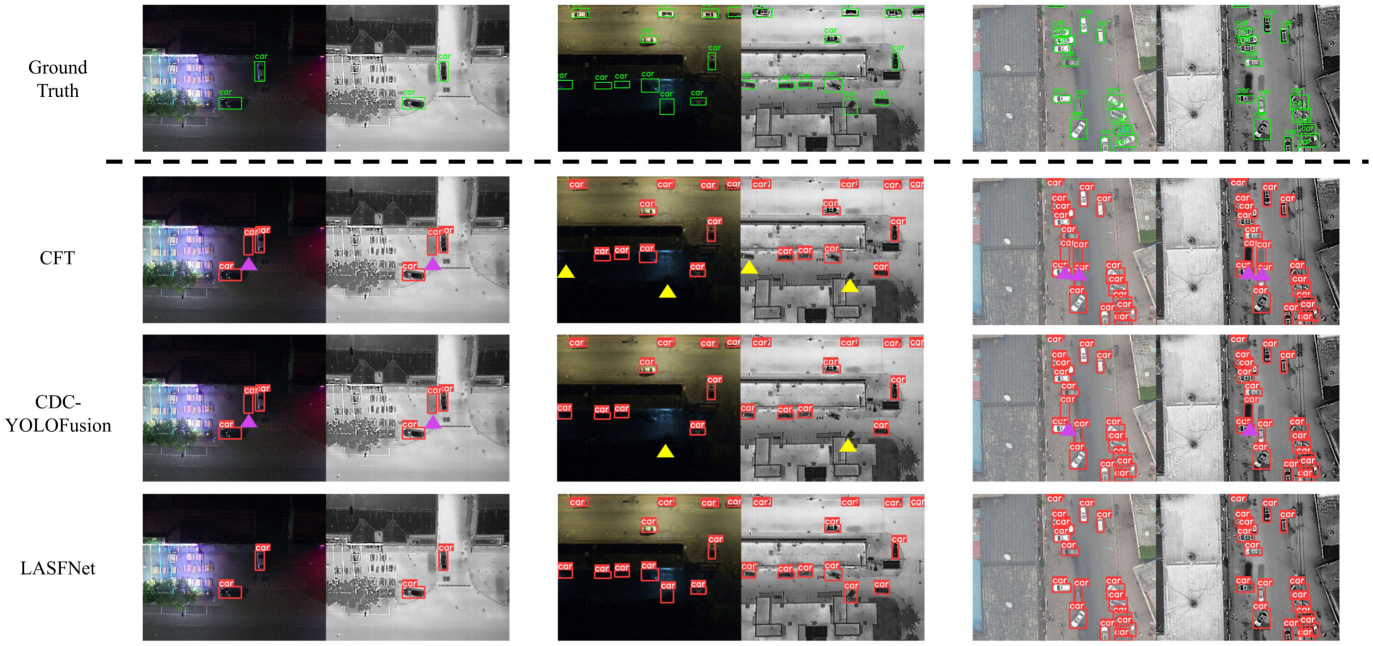


Fig. 5. Three examples on DroneVehicle dataset to illustrate the detection results, where the purple triangles represent false positive objects, and the yellow triangles indicate the missing objects.

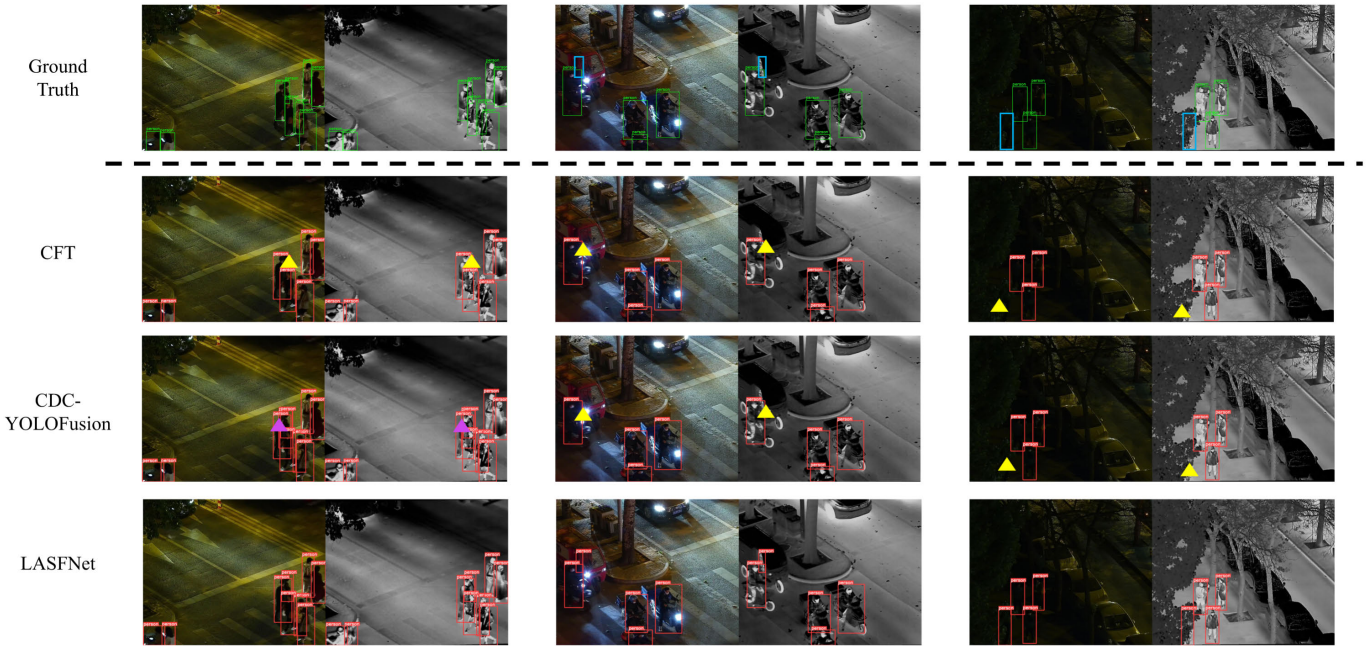


Fig. 6. Three examples on LLVIP dataset to illustrate the detection results, where the purple triangles represent false-positive objects, and the yellow triangles indicate the missing objects. In addition, the blue boxes highlight human annotation omissions in the ground truth. (a) Three examples on DroneVehicle dataset to illustrate the detection results, where the purple triangles represent false positive objects, and the yellow triangles indicate the missing objects. (b) Three examples on LLVIP dataset to illustrate the detection results, where the purple triangles represent false positive objects, and the yellow triangles indicate the missing objects. In addition, the blue boxes highlight human annotation omissions in the ground truth.

up to 90% and computational cost by up to 85%, while increasing mAP by 1%–3%. The improvements in accuracy and efficiency can be attributed to three key factors. First, we have introduced a novel and efficient lightweight fusion detection baseline. Second, the lightweight ASFF module adaptively enhances the richness and prominence of fused features through attention guidance, resulting in comprehensive and effective feature fusion. At last, the FATM module enhances the feature representation at the neck of the network in a lightweight manner, increasing the network's focus on multiscale features, reducing information loss, and improving the effectiveness of feature representation and fusion. Conse-

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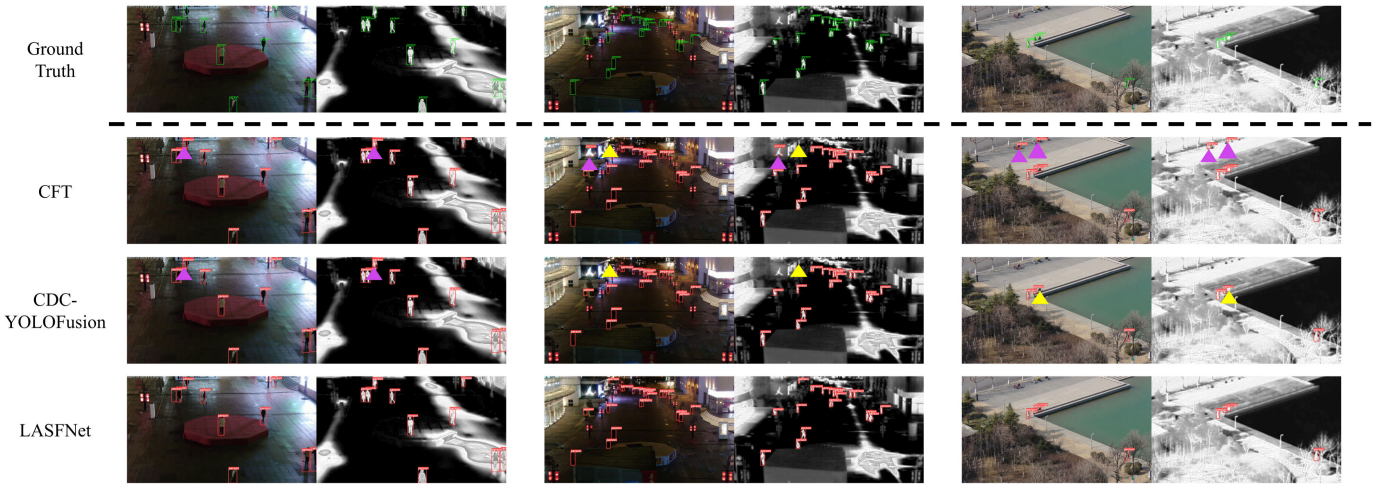


Fig. 7. Three examples on VTUAV_det dataset to illustrate the detection results, where the purple triangles represent false-positive objects, and the yellow triangles indicate the missing objects.

TABLE V
PERFORMANCE COMPARISON BETWEEN LASFNET AND SEVERAL STATE-OF-THE-ART APPROACHES ON THE DRONEVEHICLE DATASET

Method	mAP50	mAP	Params(M)	GFLOPs
(2021) CFT [59]	0.849	0.628	44.4	35.6
(2023 TGRS) SuperYOLO [36]	0.814	0.595	**4.8**	55.9
(2023 TGRS) GHOST [60]	0.814	0.594	**4.8**	54.7
(2023 RS) ADCNet [61]	0.776	0.547	25.9	40.2
(2024 PR) ICAFusion [62]	0.851	0.621	20.1	29.7
(2024 CVPR) GM-DETR [63]	0.815	0.559	70.0	176.0
(2024 TIV) CDC-YOLOFusion [37]	0.848	0.633	33.3	27.8
(2025 IF) BDDS [64]	0.838	0.618	11.4	83.0
LASFNet(Ours)	**0.857**	**0.647**	7.7	**26.6**

TABLE VI
PERFORMANCE COMPARISON BETWEEN LASFNET AND SEVERAL STATE-OF-THE-ART APPROACHES ON THE LLVIP DATASET

Method	mAP50	mAP	Params(M)	GFLOPs
(2021) CFT [59]	0.966	0.630	44.4	35.6
(2023 TGRS) SuperYOLO [36]	0.940	0.582	**4.8**	55.9
(2023 TGRS) GHOST [60]	0.939	0.586	**4.8**	54.7
(2023 RS) ADCNet [61]	0.954	0.604	25.9	40.2
(2024 PR) ICAFusion [62]	0.978	0.630	20.1	29.7
(2024 CVPR) GM-DETR [63]	0.971	0.670	70.0	176.0
(2024 TIV) CDC-YOLOFusion [37]	**0.979**	0.665	33.3	27.8
(2024) MDLNet-X [65]	0.954	0.627	189.4	1038.2
(2025 TMM) Fusion-Mamba [23]	0.968	0.628	244.6	-
LASFNet(Ours)	0.974	**0.676**	7.7	**26.6**

TABLE VII
PERFORMANCE COMPARISON BETWEEN LASFNET AND SEVERAL STATE-OF-THE-ART APPROACHES ON THE VTUAV_DET DATASET

Method	mAP50	mAP	Params(M)	GFLOPs
(2021) CFT [59]	0.760	0.343	44.4	35.6
(2023 TGRS) SuperYOLO [36]	0.773	0.345	**4.8**	55.9
(2023 TGRS) GHOST [60]	0.768	0.339	**4.8**	54.7
(2023 RS) ADCNet [61]	0.700	0.311	25.9	40.2
(2024 PR) ICAFusion [62]	0.762	0.337	20.1	29.7
(2024 CVPR) GM-DETR [63]	0.756	0.350	70.0	176.0
(2024 TIV) CDC-YOLOFusion [37]	0.788	0.357	33.3	27.8
(2024 ISPRS) QFDet [29]	0.755	0.331	60.3	242.8
(2025 TCSVT) COXNet [66]	0.761	0.335	75.9	123.6
LASFNet(Ours)	**0.797**	**0.378**	7.7	**26.6**

TABLE VIII
MODEL COMPLEXITY AND TESTING TIME COMPARISON AMONG SEVERAL MULTIMODAL DETECTION APPROACHES TESTED ON THE LLVIP DATASET

Method	Params(M)	GFLOPs	Testing time(ms)
(2021) CFT [59]	44.4	35.6	86.9
(2023 TGRS) SuperYOLO [36]	**4.8**	55.9	94.9
(2023 TGRS) GHOST [60]	**4.8**	54.7	108.1
(2023 RS) ADCNet [61]	25.9	40.2	68.4
(2024 PR) ICAFusion [62]	20.1	29.7	55.8
(2024 CVPR) GM-DETR [63]	70.0	176.0	90.6
(2024 TIV) CDC-YOLOFusion [37]	33.3	27.8	94.4
LASFNet(Ours)	7.7	**26.6**	**49.8**

quently, our approach achieves a favorable tradeoff between efficiency and accuracy.

Table VIII compares the performance of various multimodal methods in terms of model parameters, computational complexity, and inference time. Our approach employs the single feature-level fusion strategy combined with the self-modulation mechanism, achieving simultaneous improvements in model parameters and inference speed. Particularly, compared to existing lightweight strategies and advanced methods, our approach offers faster inference speeds with fewer parameters and reduced computational requirements, enabling more efficient detection.

Figs. 5–7 present qualitative detection results of CFT, CDC-YOLOFusion, and the proposed method across three datasets. Each figure displays ground truths followed by detections from the three methods. As shown in Fig. 5 on the DroneVehicle, our method accurately detects objects concealed in darkness and suppresses false alarms caused by thermal artifacts, thanks to the adaptive feature modu-

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lation in the ASFF module. In comparison, both CFT and CDC-YOLOFusion exhibit noticeable missed detections and false positives. Fig. 6 illustrates results on the LLVIP. LASFNet handles dense and heavily occluded objects more effectively, owing to the progressive complementary fusion in ASFF. The competing methods, by contrast, show increased omissions and spurious detections under these conditions. Fig. 7 demonstrates the robustness of LASFNet to multiscale variations and misalignment issues. The FATM module enhances focus on fused features and integrates multiscale contexts, leading to fewer missed detections and false positives compared to CFT and CDC-YOLOFusion.

V. Conclusion

This article presents LASFNet, a lightweight self-modulation feature fusion network for multimodal object detection. LASFNet introduces a novel single feature-level fusion unit that delivers efficient detection performance while maintaining a compact model. In implementation, we design an ASFF of the two input modality features. This effective design enables the network to progressively and adaptively adjust and enhance the fused features, providing robust support for multimodal detection. Furthermore, a FATM is incorporated into the neck of the network, which enhances the ability of the model to focus on discriminative features and improve its multiscale detection capability. Extensive experiments on three benchmark datasets demonstrate that LASFNet is both lightweight and effective, achieving a favorable tradeoff between efficiency and accuracy.

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